Pat Lank

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RESEARCH

Interests. Algebraic geometry, especially towards singularity theory, using methods from derived categories and homological algebra.

Published & accepted.

(1) *Compact approximation and descent for algebraic stacks*
- with Jack Hall, Fei Peng, Alicia Lamarche
- arXiv, accepted to **Forum Math. Sigma**

(2) *Measuring birational derived splinters*
- with Timothy De Deyn, Kabeer Manali Rahul, Sridhar Venkatesh
- arXiv, **Bull. Lond. Math. Soc.**

(3) *Integral transforms on singularity categories for Noetherian schemes*
- with Uttaran Dutta, Kabeer Manali Rahul
- arXiv, accepted to **Michigan Math. J.**

(4) *Measuring rationality of Schwede–Takagi pairs*
- with Peter McDonald, Sridhar Venkatesh
- arXiv, **J. Pure Appl. Algebra**

(5) *Triangulated characterizations of singularities*
- with Sridhar Venkatesh
- arXiv, **Nagoya Math. J.**

(6) *Closedness of the singular locus and generation for derived categories*
- with Souvik Dey
- arXiv, **J. Algebra**

(7) *Dévissage for generation in derived categories*
- with Souvik Dey
- arXiv, **Proc. Amer. Math. Soc.**

(8) *Approximation by perfect complexes detects Rouquier dimension*
- with Noah Olander
- arXiv, **Mosc. Math. J.**

(9) *Preservation for generation along the structure morphism of coherent algebras over a scheme*
- with Anirban Bhaduri, Souvik Dey
- arXiv, **Bull. Lond. Math. Soc.**

(10) *Descent conditions for generation in derived categories*
- arXiv, **J. Pure Appl. Algebra**

(11) *Strong generation for module categories*
- with Souvik Dey, Ryo Takahashi
- arXiv, **J. Pure Appl. Algebra**

(12) *High Frobenius pushforwards generate the bounded derived category*
- with Matthew Ballard, Srikanth Iyengar, Alapan Mukhopadhyay, Josh Pollitz
- arXiv, **Forum Math. Sigma**

Under revision.

(1) *Fppf descent and localizing subcategories for algebraic spaces*
- arXiv
- *European Math. J.*

(2) *Descent and generation for noncommutative coherent algebras over schemes*
- with Timothy De Deyn, Kabeer Manali Rahul, arXiv
- *J. Lond. Math. Soc.*

Selected preprints. A complete list of preprints is available on arXiv.

(1) *Germwise singular Fourier–Mukai partnerships*
- arXiv

(2) *Gluing weak approximability along recollements*
- arXiv

(3) *A blowup criterion for regularity*
- arXiv

(4) *Fourier–Mukai loci are open and base change*
- with Elías Guisado Villalgordo
- arXiv

(5) *Proxy smallness meets t-structures*
- with Michal Hrbek, Giovanna Le Gros, Sergio Pavon
- arXiv

(6) *Remarks on diagonal dimension for algebraic stacks*
- with Fei Peng
- arXiv

(7) *Dualizing complexes and t-structures for algebraic spaces*
- arXiv

(8) *Fiberwise criteria for Fourier–Mukai equivalences*
- with Elías Guisado Villalgordo, Kabeer Manali Rahul, Nebojsa Pavic
- arXiv

(9) *Nonexistence of singly compactly generated t-structures for schemes*
- with Anirban Bhaduri, Timothy De Deyn, Michal Hrbek, Kabeer Manali Rahul
- arXiv

(10) *A note on quasi-perfect morphisms*
- with Timothy De Deyn, Kabeer Manali Rahul
- arXiv

(11) *Simple criteria for higher rational singularities*
- with Sándor Kovács, Sridhar Venkatesh
- arXiv

(12) *Perfectly generated t-structures for algebraic spaces*
- with Michal Hrbek, Simone Pizzirani
- arXiv